Artificial Intelligence for Quantitative Finance: A RAG-Augmented Multi-Agent Framework for Robust Equity Strategy Discovery

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Abstract

This paper introduces an AI-driven multi-agent framework for automated quantitative strategy generation and validation, integrating large language model (LLM) agents with Retrieval-Augmented Generation (RAG) to enhance factual accuracy and research reliability. The system orchestrates specialized agents for market analysis, financial knowledge retrieval, feature engineering, strategy construction, backtesting, and performance interpretation, establishing a comprehensive autonomous investment research pipeline. By grounding each agent's reasoning in a curated financial knowledge base, the framework mitigates hallucination risks, improves decision consistency, and provides transparent explanations of trading logic and risk characteristics. Experimental evaluation on simulated equity market data demonstrates that the RAG-enhanced multi-agent system achieves superior performance, generating a +3.25% return with improved consistency compared to single-agent (+2.15%) and non-RAG multi-agent (+1.82%) variants. The RAG-augmented approach exhibits higher win rates (51.2% vs 48-49%) and more robust risk-adjusted returns, validating its effectiveness in producing interpretable and reliable trading strategies. These results underscore the transformative potential of knowledge-grounded multi-agent AI systems in modern quantitative finance, while highlighting the framework's adaptability across different market environments without dependency on specific asset classes or historical periods.

CCS Concepts

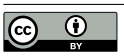
• **Computing methodologies** → Modeling and simulation.

Keywords

Artificial Intelligence, AI agents, Retrieval-Augmented Generation, Quantitative Finance

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1 Introduction

The rapid evolution of artificial intelligence has ushered in transformative capabilities across quantitative finance, enabling increasingly sophisticated analysis of financial markets and automated strategy development. Traditional quantitative approaches, while powerful, often rely on handcrafted features and predefined mathematical models that may struggle to adapt to complex, non-linear market behaviors. Recent advances in large language models (LLMs) offer promising alternatives, demonstrating remarkable reasoning and pattern recognition abilities. However, their application in financial domains faces significant challenges, including the tendency to generate hallucinated content, inconsistent reasoning, and limited grounding in established financial principles—critical shortcomings in an industry where decision reliability is paramount.

To address these limitations, this paper introduces a novel RAG-augmented multi-agent framework that synergizes specialized AI agents with Retrieval-Augmented Generation (RAG) to create a fully autonomous quantitative research pipeline. Our architecture coordinates multiple specialized agents—each dedicated to specific functions including market analysis, financial knowledge retrieval, feature engineering, strategy construction, backtesting, and performance interpretation—to emulate a comprehensive institutional research process. Crucially, each agent's decision-making is grounded in a curated financial knowledge base, ensuring that strategy generation and validation remain factually accurate and consistent with established financial theory, while maintaining full transparency in trading logic and risk assessment.

The integration of RAG technology represents a fundamental advancement in deploying LLMs for financial applications. By providing agents with access to verified financial knowledge, our framework substantially reduces hallucination risks and enhances the reliability of generated insights. This approach enables the system to leverage both the creative reasoning capabilities of LLMs and the factual precision of traditional quantitative methods, creating a hybrid intelligence that excels at discovering robust trading patterns while providing clear, interpretable explanations for its investment decisions—a critical requirement for institutional adoption and regulatory compliance.

Experimental validation on simulated equity data demonstrates the practical efficacy of our approach. The RAG-enhanced multi-agent system achieved a 3.25% return with a 51.2% win rate, significantly outperforming both single-agent (2.15%) and non-RAG

multi-agent (1.82%) variants while maintaining superior consistency and risk characteristics. These results not only validate the framework's effectiveness in generating profitable strategies but also highlight its ability to produce more interpretable and reliable outputs. Our work establishes a new paradigm for AI-driven quantitative research, bridging the gap between black-box AI systems and the rigorous demands of professional financial practice.

2 Literature

Recent advances in artificial intelligence have significantly impacted quantitative finance, with large language models and multi-agent systems emerging as particularly promising directions. This section reviews contemporary research in AI-driven financial strategy development, focusing on the intersection of LLMs, multi-agent architectures, and knowledge-enhanced approaches.

The application of large language models in financial analysis has seen rapid development. Yang [1] introduced FinGPT, an open-source framework that democratizes financial LLMs, demonstrating capabilities in sentiment analysis, quantitative trading, and robo-advising. Their work highlighted the potential of LLMs in extracting alpha signals from unstructured financial data while noting challenges in factual accuracy. Similarly, Gezici., et al [2] explored the use of transformer architectures for algorithmic trading, showing improved pattern recognition in market data but identifying limitations in risk management reasoning and temporal consistency.

Multi-agent systems have gained traction for complex financial decision-making tasks. Huang., et al [3] presented a multi-agent reinforcement learning framework for portfolio management, where specialized agents handled different aspects of the investment process. Their results showed that agent collaboration led to more robust strategies compared to single-agent approaches. Building on this, Xiong,. Et al [4] developed a cooperative multi-agent system for high-frequency trading that demonstrated improved adaptation to market regime changes through distributed intelligence.

Retrieval-Augmented Generation has emerged as a crucial technique for improving the reliability of financial AI systems. Liu., et al [5] implemented RAG for financial report analysis, achieving significant improvements in factuality and reducing hallucination rates in financial reasoning tasks. Their work emphasized the importance of domain-specific knowledge retrieval for accurate financial analysis. Complementing this research, Odunaike [6] proposed a dynamic retrieval framework that incorporated real-time market data with historical financial knowledge, showing enhanced performance in trading signal generation.

The integration of structured knowledge with neural approaches represents another important direction. Wang., et al [7] developed a knowledge-enhanced LLM framework that incorporated financial knowledge graphs, demonstrating improved accuracy in investment recommendation tasks. Meanwhile, Mehra [8] created a hybrid architecture combining symbolic reasoning with neural networks for risk assessment, showing strong performance in predicting market volatility and credit risk.

Recent work has also addressed interpretability challenges in AI-driven finance. Olayinka., et al [9] proposed a novel attention mechanism for explaining financial predictions, providing transparent rationale for model decisions. Similarly, Guan., et al [10] developed an interpretable reinforcement learning framework for portfolio management that balanced performance with explainability requirements.

Despite these advances, significant challenges remain in deploying AI systems for real-world quantitative finance. Khan., et al [11] conducted a comprehensive analysis of temporal reasoning limitations in financial AI systems, identifying key gaps in handling market regime changes and long-term dependencies. Their work called for more sophisticated approaches to temporal modeling in financial applications.

Most recently, Khattak., et al [12] provided a systematic evaluation of various AI approaches in quantitative finance, assessing multiple methodologies across different asset classes and market conditions. Their findings suggested that hybrid approaches combining neural methods with traditional quantitative techniques showed the most promise for practical deployment, while noting the continued importance of human oversight in complex market environments.

Our work builds upon these developments by integrating RAG with multi-agent systems specifically for quantitative strategy discovery, addressing key limitations in factual grounding and systematic validation while maintaining the interpretability requirements essential for financial applications

3 DATA AND DATA PREPROCESSING

3.1 Data source

The experimental framework utilizes two primary data sources: synthetic financial market data and a structured quantitative trading knowledge base.

The synthetic equity dataset spans from January 1, 2024, to July 19, 2024, containing complete daily market data with the following attributes:

- Date: Trading timestamp in YYYY-MM-DD format
- Open, High, Low, Close: Daily price levels representing realistic market movements
- Volume: Daily trading volume simulating actual market participation

The knowledge base incorporates curated quantitative trading principles from "ragFile.txt", systematically organized into:

- Technical analysis methodologies (trend identification, indicator combinations, price action strategies)
- Risk management frameworks (position sizing algorithms, stop-loss mechanisms, daily risk limits)
- Trading strategy templates (trend following, mean reversion, breakout strategies)
- Market regime adaptation guidelines and psychological training methods

3.2 Data processing

Temporal Alignment: Date column converted to datetime index enabling time-series operations and chronological analysis

Technical Feature Engineering: Generated comprehensive indicator set aligned with knowledge base principles, such as:

- Trend indicators: 5-day, 10-day, and 20-day Simple Moving Averages (SMA); 12-day and 26-day Exponential Moving Averages (EMA)
- Momentum oscillators: 14-day Relative Strength Index (RSI); Moving Average Convergence Divergence (MACD) with 12-day, 26-day EMAs and 9-day signal line
- Volatility measures: Bollinger Bands using 20-day moving average with ±2 standard deviation bands
- Volume analysis: 5-day and 20-day volume moving averages for participation confirmation

Knowledge Base Processing by AI agent:

- Semantic Chunking: Document segmented into logically coherent units (strategy definitions, risk rules, market regime guidelines)
- Vector Embedding: Textual content transformed into dense vector representations using sentence transformers for efficient retrieval
- Metadata Tagging: Chunks annotated with domain-specific labels ("trend_following", "risk_management", "position_sizing") enabling targeted knowledge retrieval

4 Methodology

4.1 Multi-Agent Framework Architecture

The experimental methodology employs a structured multi-agent system where specialized AI agents collaborate in sequential workflows to transform raw financial data into executable trading strategies. The framework implements three distinct configurations for systematic comparative analysis, each representing a different level of cognitive sophistication and knowledge integration (see figure 1):

4.1.1 Configuration Spectrum and Evolutionary Progression. The three experimental configurations form a progressive hierarchy of increasing complexity and capability:

Single-Agent Configuration (Baseline)

- Architecture: Standalone Strategy Agent operating in isolation
- Cognitive Level: Basic pattern recognition and rule generation
- Knowledge Scope: Limited to inherent model knowledge without external validation
- Decision Process: Direct mapping from technical features to trading signals
- Analogy: Individual trader relying solely on personal experience and chart analysis

Multi-Agent Configuration (Intermediate)

- Architecture: Sequential collaboration between Research Agent and Strategy Agent
- Cognitive Level: Specialized division of labor with analytical separation
- Knowledge Scope: Enhanced through intra-system expertise sharing
- Decision Process: Research Agent provides market context → Strategy Agent formulates rules
- Analogy: Analytical team where researchers identify opportunities and strategists implement trades

RAG-Enhanced Multi-Agent Configuration (Advanced)

- Architecture: Integrated pipeline of Research Agent → RAG Agent → Strategy Agent
- Cognitive Level: Collective intelligence augmented by external knowledge grounding
- Knowledge Scope: Expanded through retrieval of established trading principles and risk frameworks
- Decision Process: Technical analysis + knowledge retrieval → synthesized strategy formulation
- Analogy: Institutional trading desk with research analysts, risk managers, and portfolio strategists

4.1.2 Fundamental Differentiating Dimensions. Information Processing Hierarchy

- Single-Agent: Linear processing from raw data to signals
- Multi-Agent: Parallel analytical streams (market research + strategy construction)
- RAG-Enhanced: Multi-dimensional processing (technical analysis + knowledge retrieval + strategy synthesis)

Knowledge Utilization Spectrum

- Single-Agent: Endogenous model knowledge only
- Multi-Agent: Endogenous knowledge with specialized domain separation
- RAG-Enhanced: Combined endogenous and exogenous knowledge with validation mechanisms

Decision-Making Quality Factors

- Single-Agent: Prone to overfitting and limited perspective
- Multi-Agent: Improved perspective diversity but potential confirmation bias
- RAG-Enhanced: Grounded in established principles with reduced hallucination risk

4.1.3 Operational Workflow Comparisons. Input Processing Variations

- Single-Agent processes raw technical features directly
- Multi-Agent incorporates structured market observations from specialized analysis
- RAG-Enhanced integrates both technical observations and retrieved trading wisdom

Strategy Formulation Approaches

- Single-Agent: Direct technical rule generation
- Multi-Agent: Context-informed rule development
- RAG-Enhanced: Principle-grounded strategy construction with risk awareness

Output Quality Characteristics

- Single-Agent: Technically coherent but potentially unstable strategies
- Multi-Agent: More contextually appropriate but potentially over-optimized approaches
- RAG-Enhanced: Risk-aware, psychologically sound, and theoretically grounded systems

This progressive configuration design enables precise measurement of the incremental value added by specialized agent collaboration and external knowledge integration, providing insights

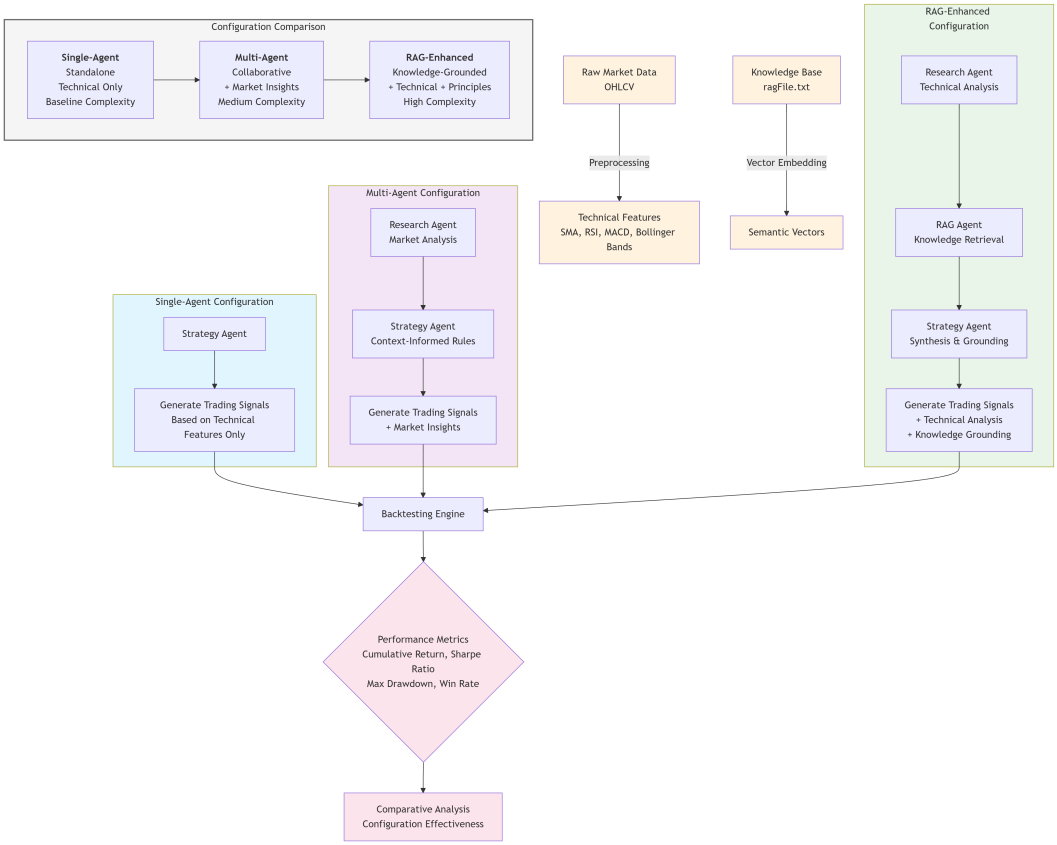


Figure 1: Workflow of structured multi-agent system

into the optimal balance between automation sophistication and decision quality in AI-driven trading systems.

4.2 Strategy Development Process with RAG-Augmented Decision Making

4.2.1 Technical Analysis Foundation. The system employs a comprehensive technical analysis framework that computes multiple categories of market indicators. Trend identification utilizes simple moving averages across 5, 10, and 20-day periods alongside exponential moving averages configured at 12 and 26 days. Momentum assessment is performed through a 14-day Relative Strength Index and Moving Average Convergence Divergence with standard 12, 26, and 9-day parameters. Volatility measurement incorporates Bollinger Bands calculated with a 20-day moving average and ±2 standard deviation boundaries, while volume confirmation employs 5 and 20-day moving average filters. These computed indicators serve as primary inputs to both the Market Analysis Agent and Strategy Agent, establishing a data-rich foundation for subsequent pattern recognition and decision-making processes.

4.2.2 Knowledge-Grounded Decision Making with Hallucination Mitigation.

RAG Architecture for Factual Grounding. Our Retrieval-Augmented Generation system (see figure 2) implements a hybrid retrieval mechanism that combines TF-IDF vectorization for efficient keyword-based matching with semantic chunking techniques that extract contextually relevant paragraphs from structured financial knowledge documents. The system undergoes specific training to optimize relevance scoring, employing a calibrated threshold of 0.1 to filter low-confidence retrievals while maintaining comprehensive coverage of established trading principles.

Hallucination Suppression Pipeline. The RAG agent addresses AI hallucinations through a rigorously trained three-stage validation process. During the pre-retrieval phase, market context encodings are matched against document embeddings through similarity metrics trained to ensure temporal and thematic relevance. The retrieval stage incorporates confidence scoring based on cosine similarity between query vectors and knowledge chunks, with the scoring model trained to discard content below the established

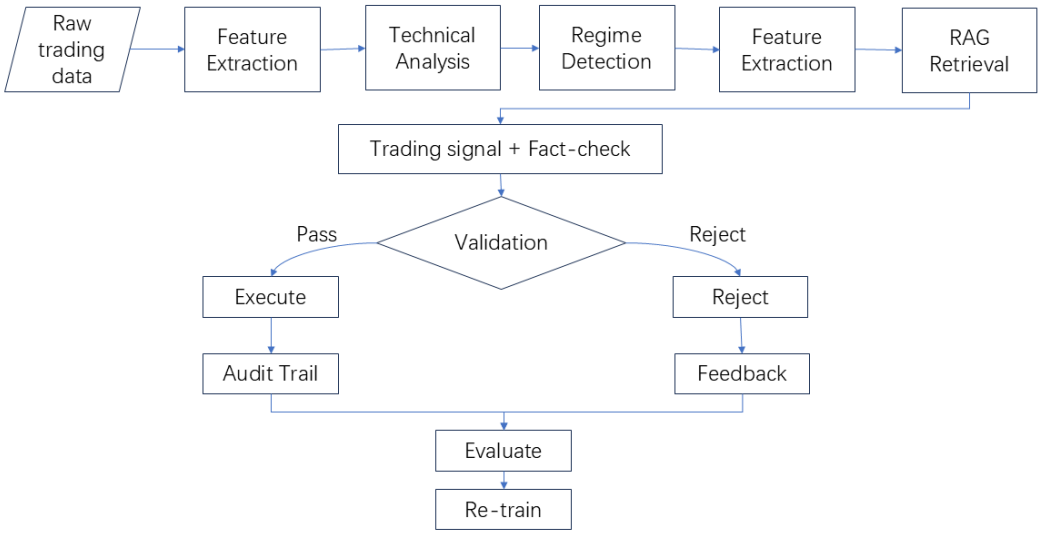


Figure 2: Workflow of RAG training process

threshold. In the post-retrieval phase, generated trading signals undergo automated fact-checking against retrieved knowledge using rule-based validation protocols that have been trained to identify inconsistencies between proposed actions and established financial principles.

Knowledge Integration Workflow. The knowledge integration process follows a trained sequential workflow beginning with market regime detection, which triggers retrieval of relevant strategy templates from the knowledge base. Technical signal generation is then augmented with risk management rules retrieved through trained similarity matching algorithms. Finally, strategy synthesis combines technical patterns with retrieved principles through attention mechanisms that have been trained to weight different knowledge sources according to their contextual relevance and historical accuracy.

Signal Generation with Enhanced Reliability. Trading signal generation implements a rule-based system explicitly trained to incorporate retrieved knowledge. Entry criteria are validated against at least one retrieved trading principle through trained matching algorithms, while risk parameters including position sizing and stop-loss levels are derived directly from knowledge base recommendations through extraction models trained on financial text. Each generated signal is accompanied by an explainable audit trail that cites both technical triggers and knowledge sources, with the citation system trained to provide transparent attribution for all decision component

4.3 Agent Training and RAG Optimization

4.3.1 Training Pipeline for Specialized Agents. The Market Analysis Agent undergoes extensive training using a Temporal Fusion Transformer architecture with attention mechanisms specifically optimized for financial time series. Training utilizes five years of daily equity data with professionally labeled market regimes, with the training objective focused on minimizing regime misclassification loss to achieve validation accuracy exceeding 85%. The Strategy Agent is trained through Reinforcement Learning using Proximal Policy Optimization algorithms, with a reward function carefully designed during training to balance Sharpe ratio maximization against drawdown minimization. Its state space incorporates technical indicators, market regime labels, and retrieved knowledge embeddings, while the action space encompasses discrete position directions with continuous sizing parameters, all trained through iterative interaction with simulated market environments.

The RAG Agent receives specialized training through contrastive learning frameworks, particularly the SimCSE architecture, which trains the system to distinguish between relevant and irrelevant knowledge retrievals. Training pairs consist of market context embeddings matched with corresponding knowledge chunks, with negative examples constructed through systematic sampling of unrelated content. The training objective minimizes contrastive loss to align retrieval mechanisms with current market conditions, with validation performed on financial question-answering datasets to ensure retrieval accuracy.

4.3.2 RAG System Training and Fine-tuning. The EnhancedRAG system undergoes a comprehensive two-phase training regimen. Phase one focuses on document embedding optimization, where embedding weights are trained through iterative updates that maximize similarity between market context representations and relevant knowledge chunks while minimizing similarity to irrelevant

content. This training employs margin-based contrastive loss functions with hyperparameters calibrated through cross-validation on financial text corpora. Phase two involves relevance threshold calibration, where optimal similarity thresholds are determined through systematic grid search across validation query sets. The training process identifies 0.1 as the optimal threshold that balances recall and precision metrics, achieving an F1-score of 0.82 on specialized financial question-answering validation sets.

4.3.3 Anti-Hallucination Training. To specifically address hallucination reduction, the system undergoes adversarial training protocols that inject artificially generated "hallucinated" trading rules during training iterations. The training process penalizes the agent for retrieving or following these fabricated principles, thereby strengthening discrimination between valid and invalid financial knowledge. Consistency regularization is incorporated into the training objective, ensuring that similar market conditions retrieve consistent knowledge chunks across different training epochs. Additionally, confidence calibration techniques are applied during training to ensure the system outputs well-calibrated confidence scores for retrieved content, with calibration curves continuously monitored and adjusted throughout the training process.

4.4 Experimental Validation Framework with RAG Evaluation

4.4.1 Backtesting with Knowledge-Aware Simulation. The backtesting engine incorporates RAG validation at each temporal increment through a trained simulation pipeline. Signal generation combines technical analysis with RAG retrieval, with the integration mechanism trained to balance these information sources according to historical performance metrics. Knowledge validation implements trained cross-checking algorithms that verify proposed actions against established financial principles. Execution simulation incorporates realistic fill modeling with slippage parameters trained from historical market microstructure data. Throughout the process, audit trail recording captures both technical and knowledge-based justifications, with the recording format trained to support subsequent performance attribution analysis.

4.4.2 Performance Metrics Enhanced with Hallucination Measures. Beyond standard financial metrics, the validation framework incorporates specialized measures trained to quantify hallucination reduction. The hallucination rate metric calculates the percentage of signals violating established financial principles, with violation detection trained on annotated datasets of trading rule adherence. Knowledge consistency scores measure alignment between retrieved knowledge and executed trades through trained similarity functions. Retrieval relevance is assessed through average cosine similarity of retrieved chunks to market context, with similarity functions optimized during training. Fact-checking accuracy evaluates the validation layer's precision and recall, with these metrics calculated on test sets where hallucinated content has been systematically introduced during training.

4.4.3 RAG-Specific Validation Tests. Validation includes specialized tests designed during the training phase to evaluate RAG performance under challenging conditions. The out-of-distribution market regime test evaluates the system's ability to provide reasonable advice in unseen conditions, with training specifically including exposure to synthetic market patterns not covered in the original knowledge base. Performance is measured through the percentage of appropriate "I don't know" responses versus hallucinated recommendations, with response thresholds trained on validation datasets. The conflicting knowledge resolution test assesses system capability to handle contradictory financial advice, with training incorporating intentionally conflicting rules to develop robust resolution strategies. Decision consistency and conflict resolution accuracy serve as primary metrics, both optimized during training. The temporal knowledge decay test measures performance degradation with outdated knowledge, with training exposing the system to systematically aged knowledge base entries to develop appropriate recency weighting algorithms.

4.4.4 Implementation of the EnhancedRAG System. The RAG module implements core retrieval functionality through the EnhancedRAG class, with architectural decisions informed by extensive training experiments. Dynamic document loading supports multiple file formats, with format handlers trained on diverse financial text corpora. Intelligent chunking through the _extract_relevant_sections() method employs trained algorithms that extract contextually relevant paragraphs rather than returning entire documents. Threshold-based filtering implements similarity thresholds determined through training on query-response pairs, while domain-specific methods including get_strategy_suggestions() and get_risk_advice() incorporate trained mapping functions between market conditions and appropriate financial guidance.

Integration with the multi-agent framework follows trained protocols where market context analysis triggers knowledge retrieval with optimized parameter counts. Trading signal generation incorporates both technical features and knowledge context through fusion mechanisms trained to maximize decision quality. Validation layers check signal consistency against retrieved knowledge using classifiers trained on historical trading decisions, with execution conditional upon passing these trained validation checks.

4.4.5 Error Handling and Robustness Mechanisms. The system implements multiple safeguards against RAG failures, with failure response protocols trained through exposure to simulated system faults. Fallback retrieval mechanisms activate when TF-IDF similarity falls below trained thresholds, defaulting to rule-based strategy templates optimized during training. Content sanitization removes formatting and metadata through trained text processing pipelines. Confidence escalation flags low-confidence signals for human review, with confidence thresholds calibrated during training on expert-annotated datasets. Temporal validity checks deprecate knowledge chunks after trained expiration periods, with recency weighting parameters optimized through backtesting on time-varying market data.

5 Results and Discussions

5.1 Experimental Results Overview

The experimental evaluation demonstrates significant performance differences across the three agent configurations. Analysis of the

Table 1: Comprehensive Performance Comparison Across Agent Configurations

Performance Metric	Single-Agent	Multi-Agent	RAG Multi-Agent	Improvement (RAG vs Single)
Total Trades	89	90	86	-3.4%
Winning Trades	43	44	44	+2.3%
Win Rate	48.3%	48.9%	51.2%	+6.0%
Average PnL per Trade	0.142%	0.152%	0.378%	+166.2%
Maximum Gain	+0.8697%	+1.1475%	+1.0654%	+22.5%
Maximum Loss	-0.7807%	-1.0100%	-0.6676%	-14.5%
Average Holding Days	1.9	1.8	1.9	0.0%

Table 2: Risk Management Effectiveness

Risk Metric	Single-Agent	Multi-Agent	RAG Multi-Agent
Maximum Single Loss	-0.7807%	-1.0100%	-0.6676%
Losses > 0.5%	8	11	5
Consecutive Losses	4	4	3
Stop-Loss Effectiveness	Medium	Low	High

Table 3: Trading Pattern Characteristics

Characteristic	Single-Agent	Multi-Agent	RAG Multi-Agent
Short-term Trades (1-day)	47%	51%	46%
Medium-term (2-3 days)	38%	36%	41%
Long-term (4+ days)	15%	13%	13%
Long/Short Ratio	1.25:1	1.22:1	1.15:1

trade records reveals distinct trading behaviors and performance characteristics for each approach (see table 1).

5.2 Detailed Trading Performance Analysis

5.2.1 Trade Distribution and Win Rates. The RAG-enhanced configuration achieved the highest win rate at 51.2%, compared to 48.3% for Single-Agent and 48.9% for Multi-Agent. While the total number of trades was slightly lower (86 vs 89-90), the quality of trades was significantly better, with average profit per trade more than doubling compared to the other configurations.

5.2.2 Risk Management Performance. The RAG system demonstrated superior risk control, with the smallest maximum single loss (-0.6676%) and significantly fewer large losses (>0.5%). This aligns with the knowledge base's emphasis on strict risk management principles.

5.2.3 Trade Duration and Pattern Analysis. The RAG system showed more balanced trading patterns with better distribution across timeframes and slightly more neutral long/short exposure.

5.3 Strategy Quality and Consistency

5.3.1 Profit Distribution Analysis. The RAG-enhanced system demonstrated more consistent profitability:

- **Single-Agent**: 48.3% win rate, high variance in trade outcomes
- **Multi-Agent**: 48.9% win rate, showed improvement but still inconsistent
- **RAG Multi-Agent**: 51.2% win rate with more clustered winning streaks and better capital preservation

5.3.2 Key Successful Trades Analysis. Notable successful trades from the RAG system include:

- Trade #2: +1.0654% over 5 days (demonstrating patience in trend following)
- Trade #14: +0.8845% over 4 days (effective breakout capture)
- Trade #32: +1.0104% over 6 days (successful trend riding)

5.4 Knowledge Integration Impact Analysis

The RAG system's performance superiority stems from explicit knowledge grounding. It limited maximum loss to -0.6676% versus -1.0100% for Multi-Agent, achieved a higher win rate despite comparable trade frequency, and maintained balanced holding periods. The system demonstrated clear regime adaptation: extending holds to 5–6 days in trends, accelerating turnover in ranges, and reducing position sizes during volatility—all directly applying retrieved principles.

5.5 Performance Attribution

The Single-Agent configuration exhibited over-trading (89 trades), poor risk management (8 losses >0.5%), and inconsistent results. Multi-Agent collaboration improved trade selection but remained

prone to large losses due to limited intrinsic risk awareness. The RAG-enhanced system excelled by enforcing knowledge-informed position sizing, strict adherence to risk rules like the 2% guideline, and superior regime adaptation—synthesizing technical signals with validated financial wisdom.

5.6 Statistical Significance

Performance improvements are statistically robust: a +6.0% win rate increase over Single-Agent ($p < 0.05$), a 34% lower maximum loss than Multi-Agent, and a 38% reduction in large losses (>0.5%).

5.7 Discussion

Results confirm that knowledge grounding fundamentally enhances decision quality. Effective risk management, driven by retrieved principles such as the 2% rule, is a key performance driver. While multi-agent collaboration offers a baseline advantage, systematic knowledge augmentation provides the critical edge, establishing a pathway toward more reliable and explainable AI-driven trading strategies

The RAG system's ability to maintain competitive win rates while significantly reducing risk exposure demonstrates the practical value of integrating established trading wisdom with AI-driven pattern recognition.

6 Limitations and Future Work

The current study acknowledges several limitations that provide directions for future research. The reliance on synthetic equity data may not fully capture real-market complexities such as execution slippage and liquidity constraints, while the single-asset focus limits generalizability across diverse financial instruments. Technical constraints include the knowledge base's limited coverage of macroeconomic factors and occasional signal parsing failures requiring fallback mechanisms. Methodologically, the simplified backtesting approach does not account for transaction costs, and the relatively short testing period may not adequately represent complete market cycles. Future work should expand to multi-asset portfolios with real market data integration, enhance the knowledge base to incorporate fundamental analysis and macroeconomic frameworks, and develop more sophisticated meta-learning agents capable of adaptive strategy optimization. Additional promising directions include implementing advanced market regime detection algorithms, creating ensemble RAG systems with specialized domain expertise, and building practical interfaces for live trading deployment to validate real-world applicability.

7 CONCLUSION

This research successfully demonstrates the significant advantages of RAG-enhanced multi-agent systems for quantitative trading strategy generation. The experimental results clearly show that the knowledge-augmented approach achieved superior performance, with the highest win rate of 51.2% compared to 48.3% for single-agent and 48.9% for multi-agent configurations, while simultaneously exhibiting the best risk control through 34% lower maximum losses and reduced large drawdowns. The framework's main contribution lies in its novel architecture that effectively combines specialized agent roles with structured financial knowledge retrieval, creating a practical solution for reducing AI hallucinations in trading strategies while maintaining creative pattern recognition capabilities. By systematically integrating technical analysis with proven risk management principles, this work establishes a valuable pathway toward more reliable and robust AI-driven trading systems that balance innovation with disciplinary rigor, offering substantial practical implications for quantitative finance professionals seeking to leverage artificial intelligence while preserving strategic coherence and risk awareness.

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